Brief Contents

Part I Introduction 1

1 Introduction to Scientific Research 1

2 Research Approaches and Methods of Data Collection 18

Part II Planning the Research Study 42

3 Identifying Research Questions and Forming Hypotheses 42

4 Ethics 60

Part III Foundations of Research 87

5 Measuring Variables and Sampling 87

6 Research Validity 109

7 Control Techniques in Research 131

8 Procedure for Conducting a Research Study 149

Part IV Experimental Methods 165

9 Experimental Research Design 165

10 Quasi-Experimental Designs 193

11 Single-Case Research Designs 210

Part V Nonexperimental, Qualitative, and Mixed Methods Research 228

12 Nonexperimental Quantitative Research Methods 228

13 Survey Research 247

14 Qualitative and Mixed Methods Research 270

Part VI Analyzing and Interpreting Data 295

15 Descriptive Statistics 295

16 Inferential Statistics 322

Part VII Writing the Research Report 354

17 Preparing the Research Report Using APA Style 354

Appendix 379

Glossary 380

References 392

Index 401

This page intentionally left blank

Contents

Preface xi

PART I Introduction 1

1 Introduction to Scientific Research 1

Traditional Methods of Acquiring Knowledge 2
- Authority 3
- Rationalism 3
- Empiricism 3

The Scientific Approach to Knowledge Generation 4
- Induction and Deduction 4
- Hypothesis Testing 5
- Science in the 21st Century 6

Assumptions Underlying Scientific Research 7
- Regularity in Nature 7
- Reality in Nature 7
- Discoverability 8

Characteristics of Scientific Research 8
- Empirical 8
- Control 9
- Operationalization of Constructs 9
- Replication 10
- Evidence Not Proof 10

The Role of Theory in Scientific Research 11

Objectives of Psychological Research 13
- Description 13
- Explanation 13
- Prediction 13
- Control or Influence 14

Pseudoscience 14

Summary 15 • Key Terms and Concepts 16 • Related Internet Sites 16 • Practice Test 16 • Challenge Exercises 17

2 Research Approaches and Methods of Data Collection 18

Variables in Quantitative Research 19

Experimental Research 21

Causation 22
- Cause 22
- Effect 23
- Required Criteria for Making the Claim of Causation 23

The Psychological Experiment 24
- Example of an Experiment and Its Logic 24
- Experimental Research Settings 26
- Advantages of Experimental Research 26
  - 1. Causal Inference 26 • 2. Ability to Manipulate Variables 27 • 3. Control over Extraneous Variables 27
- Disadvantages of Experimental Research 27
  - 1. Does Not Test Effects of Nonmanipulated Variables 27 • 2. Artificiality 27 • 3. Incomplete Method of Scientific Inquiry 27

Nonexperimental Quantitative Research 28
- Advantages and Disadvantages of Nonexperimental Quantitative Research 30

Qualitative Research 31

Mixed Methods Research 33

Major Methods of Data Collection 33
- Tests 34
- Questionnaires 34
- Interviews 35
- Focus Groups 35
- Observation 36
- Existing or Secondary Data 37

Summary 39 • Key Terms and Concepts 40 • Related Internet Sites 40 • Practice Test 40 • Challenge Exercises 41

PART II Planning the Research Study 42

3 Identifying Research Questions and Forming Hypotheses 42

Sources of Research Ideas 43
- Everyday Life 43
- Practical Issues 43
- Past Research 44
- Theory 44

Breadth of Research Ideas 45
- Ideas Not Capable of Scientific Investigation 46

Review of the Literature 46
- Getting Started 47
- Defining Objectives 47
- Conducting the Search 47
  - Books 48 • Psychological Journals 48 • Computerized or Electronic Databases 48 • Internet Resources 51
- Obtaining Resources 53
- Additional Information Sources 53

Feasibility of the Study 54

Purpose of the Research 55
- Defining the Research Question 55
- Specificity of the Research Question 55

Formulating Hypotheses 56

Summary 57 • Key Terms and Concepts 58 • Related Internet Sites 58 • Practice Test 58 • Challenge Exercises 59

4 Ethics 60

Research Ethics: What Are They? 61
Relationship Between Society and Science 61
Professional Issues 62
Treatment of Research Participants 64
Ethical Dilemmas 64
Ethical Guidelines 66
Beneficence and Nonmaleficence 67
Fidelity and Responsibility 67
Integrity 68
Justice 68
Respect for People's Rights and Dignity 68
APA Ethical Standards for Research 68
Institutional Approval 69
Informed Consent 70
Informed Consent and Minors 70 • Passive Versus Active Consent 71
Deception 72
Debriefing 73
Coercion and Freedom to Decline Participation 74
Confidentiality, Anonymity, and the Concept of Privacy 75
Ethical Issues in Electronic Research 76
Informed Consent and Internet Research 76
Privacy and Internet Research 77
Debriefing and Internet Research 77
Ethical Issues in Preparing the Research Report 78
Authorship 78
Writing the Research Report 79
Ethics of Animal (Nonhuman) Research 80
Safeguards in the Use of Animals 80
Animal Research Guidelines 80
I. Justification of the Research 80 • II. Personnel 81 • III. Care and Housing of Animals 81 • IV. Acquisition of Animals 81 • V. Experimental Procedures 81 • VI. Field Research 82 • VII. Educational Use of Animals 82
Summary 83 • Key Terms and Concepts 84 • Related Internet Sites 85 • Practice Test 85 • Challenge Exercises 85

PART III Foundations of Research 87

5 Measuring Variables and Sampling 87

Defining Measurement 88
Scales of Measurement 88
Nominal Scale 88 • Ordinal Scale 89 • Interval Scale 89 • Ratio Scale 89
Psychometric Properties of Good Measurement 90
Overview of Reliability and Validity 90
Reliability 90
Test–Retest Reliability 90 • Equivalent-Forms Reliability 90 • Internal Consistency Reliability 91 • Interrater Reliability 91
Validity 91
Validity Evidence Based on Content 92 • Validity Evidence Based on Internal Structure 93 • Validity Evidence Based on Relations to Other Variables 93
Using Reliability and Validity Information 95
Sources of Information About Tests 95
Sampling Methods 95
Terminology Used in Sampling 96
Random Sampling Techniques 97
Simple Random Sampling 97 • Stratified Random Sampling 99 • Cluster Random Sampling 101 • Systematic Sampling 101
Nonrandom Sampling Techniques 102
Random Selection and Random Assignment 103
Determining the Sample Size When Random Sampling Is Used 104
Sampling in Qualitative Research 105
Summary 106 • Key Terms and Concepts 107 • Related Internet Sites 107 • Practice Test 108 • Challenge Exercises 108

6 Research Validity 109

Overview of Four Major Types of Validity 110
Statistical Conclusion Validity 110
Construct Validity 111
Threats to Construct Validity 112
Participant Reactivity to the Research Situation 112 • Researcher Effects 113
Internal Validity 115
Threats to Internal Validity 116
History 118 • Maturation 120 • Instrumentation 120 • Testing 120 • Regression Artifact 121 • Attrition 121 • Selection 122 • Additive and Interactive Effects 122
External Validity 123
Population Validity 124
Ecological Validity 125
Temporal Validity 125
Treatment Variation Validity 125
Outcome Validity 126
Relationship Between Internal and External Validity 126
Summary 127 • Key Terms and Concepts 128 • Related Internet Sites 128 • Practice Test 128 • Challenge Exercises 129

7 Control Techniques in Research 131

Control Techniques Carried Out at the Beginning of the Study 133
Random Assignment 133
Matching 136
Matching by Holding Variables Constant 136
Matching by Equating Participants 137
Building the Extraneous Variable into the Design 138
Statistical Control 139

Control Techniques Carried Out During the Study 139
Counterbalancing 139
Randomized Counterbalancing 141
Complete Counterbalancing 141
Incomplete Counterbalancing 142
Blind Technique 143
Double-Blind Technique 144
Single-Blind Technique 144
Partial-Blind Technique 144
Automation 145
Likelihood of Achieving Control 145
Detecting Participant Interpretation 145
Summary 146 • Key Terms and Concepts 147 • Related Internet Sites 147 • Practice Test 147 • Challenge Exercises 148

8 Procedure for Conducting a Research Study 149

Institutional Approval 150
Research Participants 151
Obtaining Animals 151
Obtaining Human Participants 151
Sample Size 153
Power 153
Apparatus and/or Instruments 156
Procedure 157
Scheduling of Research Participants 157
Consent to Participate 158
Instructions 159
Data Collection 159
Debriefing 159
Debriefing Functions 159 • How to Debrief 160
Pilot Study 161
Summary 162 • Key Terms and Concepts 163 • Related Internet Site 163 • Practice Test 163 • Challenge Exercise 164

PART IV Experimental Methods 165

9 Experimental Research Design 165

Weak Experimental Research Designs 167
One-Group Posttest-Only Design 168
One-Group Pretest–Posttest Design 168
Posttest-Only Design with Nonequivalent Groups 169
Strong Experimental Research Designs 170
Between-Participants Designs 172
Posttest-Only Control-Group Design 172
Strengths and Weaknesses of the Posttest-Only Control-Group Design 173
Within-Participants Designs 175
Strengths and Weaknesses of Within-Participants Designs 175
Mixed Designs (i.e., Combination of Between and Within) 176
Pretest–Posttest Control-Group Design 177
Advantages and Disadvantages of Including a Pretest 177
Factorial Designs 179
Factorial Designs Based on Within-Participants Independent Variables 184
Factorial Designs Based on a Mixed Model 185
Nature of Independent Variables in Factorial Designs 186
Strengths and Weaknesses of Factorial Designs 186
How to Choose or Construct the Appropriate Experimental Design 188
Summary 189 • Key Terms and Concepts 190 • Related Internet Sites 190 • Practice Test 190 • Challenge Exercises 191

10 Quasi-Experimental Designs 193

Quasi-Experimental Designs 194
Nonequivalent Comparison Group Design 196
Outcomes with Rival Hypotheses 198
Outcome I: Increasing Control and Experimental Groups 198 • Outcome II: Experimental-Group-Higher-than-Control-Group -at-Pretest Effect 199 • Outcome III: Experimental-Group-Lower-than-Control-Group-at-Pretest Effect 199 • Outcome IV: Crossover Effect 200
Ruling Out Threats to the Nonequivalent Comparison Group Design 200
Causal Inference from the Nonequivalent Comparison Group Design 202
Time-Series Design 202
Interrupted Time-Series Design 203
Regression Discontinuity Design 204
Summary 206 • Key Terms and Concepts 207 • Related Internet Sites 207 • Practice Test 207 • Challenge Exercises 208

11 Single-Case Research Designs 210

History of Single-Case Designs 211
Single-Case Designs 213
ABA and ABAB Designs 214
Combination Design 216
Multiple-Baseline Design 217
Changing-Criterion Design 218
Methodological Considerations in Using Single-Case Designs 220
Baseline 220
Changing One Variable at a Time 221
Length of Phases 222
Criteria for Evaluating Change 222
Experimental Criterion 223
Therapeutic Criterion 224

Rival Hypotheses 224
Summary 225 • Key Terms and Concepts 226 • Related Internet Sites 226 • Practice Test 226 • Challenge Exercises 227

PART V Nonexperimental, Qualitative, and Mixed Methods Research 228

12 Nonexperimental Quantitative Research Methods 228

Independent Variables and Dependent Variables in Nonexperimental Quantitative Research 229
The Bradford Hill Criteria for Causation 230
Techniques of Control in Nonexperimental Quantitative Research 232
- Matching 232
- Holding the Extraneous Variable Constant 234
- Statistical Control 234

Nonexperimental Designs Based on the Time Dimension 235
- Cross-Sectional and Longitudinal Designs 235
- Evaluation of Cross-Sectional and Longitudinal Designs 237

Nonexperimental Research Designs based on Research Purpose 238
- Descriptive Nonexperimental Quantitative Research 238
- Predictive Nonexperimental Quantitative Research 239
- Explanatory Nonexperimental Quantitative Research 240

Summary 244 • Key Terms and Concepts 245 • Related Internet Sites 245 • Practice Test 245 • Challenge Exercises 246

13 Survey Research 247

Purposes of Survey Research 249
Steps in Survey Research 250
Cross-Sectional and Longitudinal Designs 251
Selecting a Survey Data Collection Method 252
Constructing and Refining a Survey Instrument 254
- Principle 1. Write Items to Match the Research Objectives 255
- Principle 2. Write Items That Are Appropriate and Meaningful for the Respondents to Be Surveyed 255
- Principle 3. Write Short, Simple Questions 256
- Principle 4. Avoid Loaded and Leading Questions 256
- Principle 5. Avoid Double-Barreled Questions 256
- Principle 6. Avoid Double Negatives 257
- Principle 7. Determine Whether Closed-Ended and/or Open-Ended Questions Are Needed 257
- Principle 8. Construct Mutually Exclusive and Exhaustive Response Categories for Closed-Ended Questions 258
- Principle 9. Consider the Different Types of Closed-Ended Response Categories Rating Scales 258
  - Binary Forced Choice 260 • Rankings 260 • Checklists 261
- Principle 10. Use Multiple Items to Measure Complex or Abstract Constructs 261
  - Semantic Differential 261 • Likert Scaling 262
- Principle 11. Make Sure the Questionnaire Is Easy to Use from the Beginning to the End 263
  - Ordering of Questions 263 • Contingency Questions 263 • Questionnaire Length 264 • Response Bias 264
- Principle 12. Pilot Test the Questionnaire Until It Is Perfected 265

Selecting Your Survey Sample from the Population 265
Preparing and Analyzing Your Survey Data 267
Summary 268 • Key Terms and Concepts 268 • Related Internet Sites 268 • Practice Test 269 • Challenge Exercises 269

14 Qualitative and Mixed Methods Research 270

Major Characteristics of Qualitative Research 272
Research Validity in Qualitative Research 273
- Descriptive Validity 274
- Interpretive or Emic Validity 274
- Theoretical Validity 274
- Internal (Causation) Validity 274
- External (Generalizing) Validity 275

Four Major Qualitative Research Methods 276
- Phenomenology 276
  - Phenomenological Data Collection and Data Analysis 277 • Phenomenological Report Writing 278
- Ethnography 279
  - Ethnographic Data Collection Methods 280 • Entry, Group Acceptance, and Fieldwork 280 • Data Analysis and Report Writing 281
- Case Study Research 282
  - Data Collection in Case Study Research 282 • Case Study Designs 282 • Case Study Data Analysis and Report Writing 283
- Grounded Theory 283
  - Data Collection in Grounded Theory Research 285 • Grounded Theory Data Analysis and Report Writing 285

Mixed Methods Research 286
Research Validity in Mixed Methods Research 287
Mixed Methods Designs 289
Summary 292 • Key Terms and Concepts 293 • Related Internet Sites 293 • Practice Test 294 • Challenge Exercises 294

PART VI Analyzing and Interpreting Data 295

15 Descriptive Statistics 295

Descriptive Statistics 296
Frequency Distributions 298
Graphic Representations of Data 299
- Bar Graphs 299
- Histograms 300
- Line Graphs 300
- Scatterplots 302

Measures of Central Tendency 304
Mode 304
Median 304
Mean 304
Measures of Variability 305
Range 306
Variance and Standard Deviation 306
Standard Deviation and the Normal Curve 306
z scores 307
Examining Relationships Among Variables 309
Unstandardized and Standardized Difference Between Group Means 309
Correlation Coefficient 310
Partial Correlation Coefficient 315
Regression Analysis 315
Contingency Tables 318
Summary 320 • Key Terms and Concepts 320 • Related Internet Sites 320 • Practice Test 321 • Challenge Exercises 321

16 Inferential Statistics 322

Sampling Distributions 323
Estimation 325
Hypothesis Testing 327
Directional Alternative Hypotheses 331
Review of the Logic of Hypothesis Testing 332
Hypothesis-Testing Errors 332
Hypothesis Testing in Practice 334
The *t* Test for Correlation Coefficients 335
One-Way Analysis of Variance 336
Post Hoc Tests in Analysis of Variance 337
Analysis of Covariance 338
Two-Way Analysis of Variance 340
One-Way Repeated Measures Analysis of Variance 342
The *t* Test for Regression Coefficients 344
Nonparametric Statistics 346
Chi-Square Test for Independence 347
Other Significance Tests 348
Hypothesis Testing and Research Design 348
Summary 350 • Key Terms and Concepts 351 • Related Internet Sites 352 • Practice Test 352 • Challenge Exercises 352

PART VII Writing the Research Report 354

17 Preparing the Research Report Using APA Style 354

The APA Format 356
Preparation of the Research Report 363
Writing Style 364
Language 365
Specificity 365 • Labels 365 • Participation 366 • Specific Issues 366
Editorial Style 367
Italics 367 • Abbreviations 367 • Headings 367 • Quotations 368 • Numbers 368 • Physical Measurements 368 • Presentation of Statistical Results 368 • Tables 368 • Figures 369 • Figure Legends and Captions 370 • Figure Preparation 370 • Reference Citations 370 • Reference List 371 • Preparation of the Manuscript for Submission 372 • Ordering of Manuscript Pages 372
Submission of the Research Report for Publication 373
Acceptance of the Manuscript 374
Presenting Research Results at Professional Conferences 374
Oral Presentation 375
Poster Presentation 375
Summary 377 • Key Terms and Concepts 377 • Related Internet Sites 378 • Practice Test 378 • Challenge Exercises 378

Appendix 379
Glossary 380
References 392
Index 401

This page intentionally left blank